Color-pattern diversity, *with and without history*

Demo 8 · pavo + phylogeny + measurement error

EEB 187 · Wednesday May 20, 2026

Today's question, in three parts

Are your family's color-pattern variables…

1. structured?
2. structured because of phylogeny?
3. structured robust to measurement noise?

Each part builds on the one before.

Last week — Frédérich et al. 2026

the result this demo benchmarks against

$$K = 0.10$$

Chaetodontidae · multivariate Blomberg's K · 30 binary motifs

Color patterns evolve fast.

Convergent across the tree.

A landmark — not a target. Your per-variable Ks won't hit 0.10.

Today's data (walkthrough)

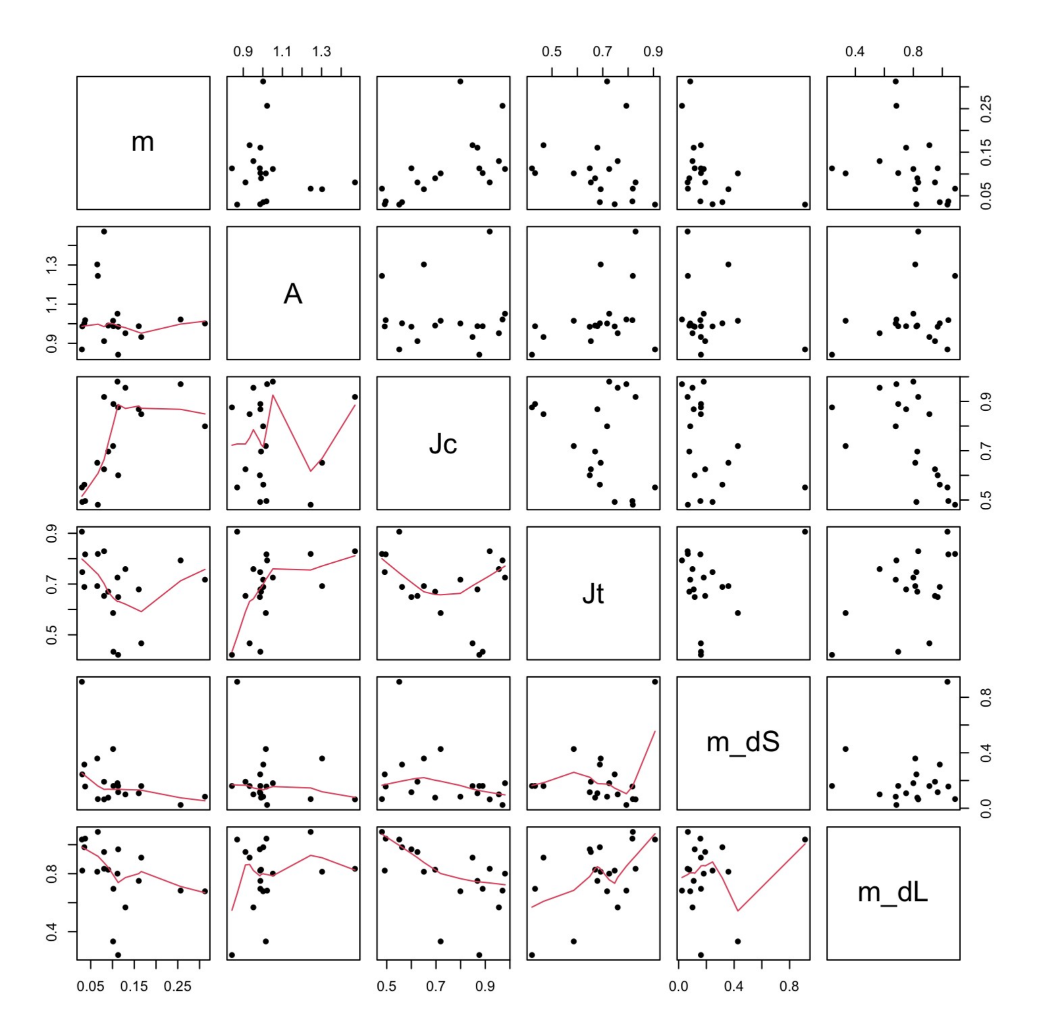
Chaetodontidae mini-bundle — pre-built, watch-only

- 19 butterflyfish species, 5 genera, matched to a pruned chronogram
- One error-species, photographed 5 times — the noise stack for Part 3
- 9 pavo pattern statistics per species
- 4 FishBase traits: trophic level, max depth, max length, vulnerability

After Part 1, every group builds a bundle of this shape for their own family.

Part 1 — describe what you've got

6 pattern stats per species · pavo classify() + adjacent()



Diagonal: distribution

Off-diagonal: scatter + smoother

Six full-rank descriptors:

m, A, Jc, Jt, m_dS, m_dL.

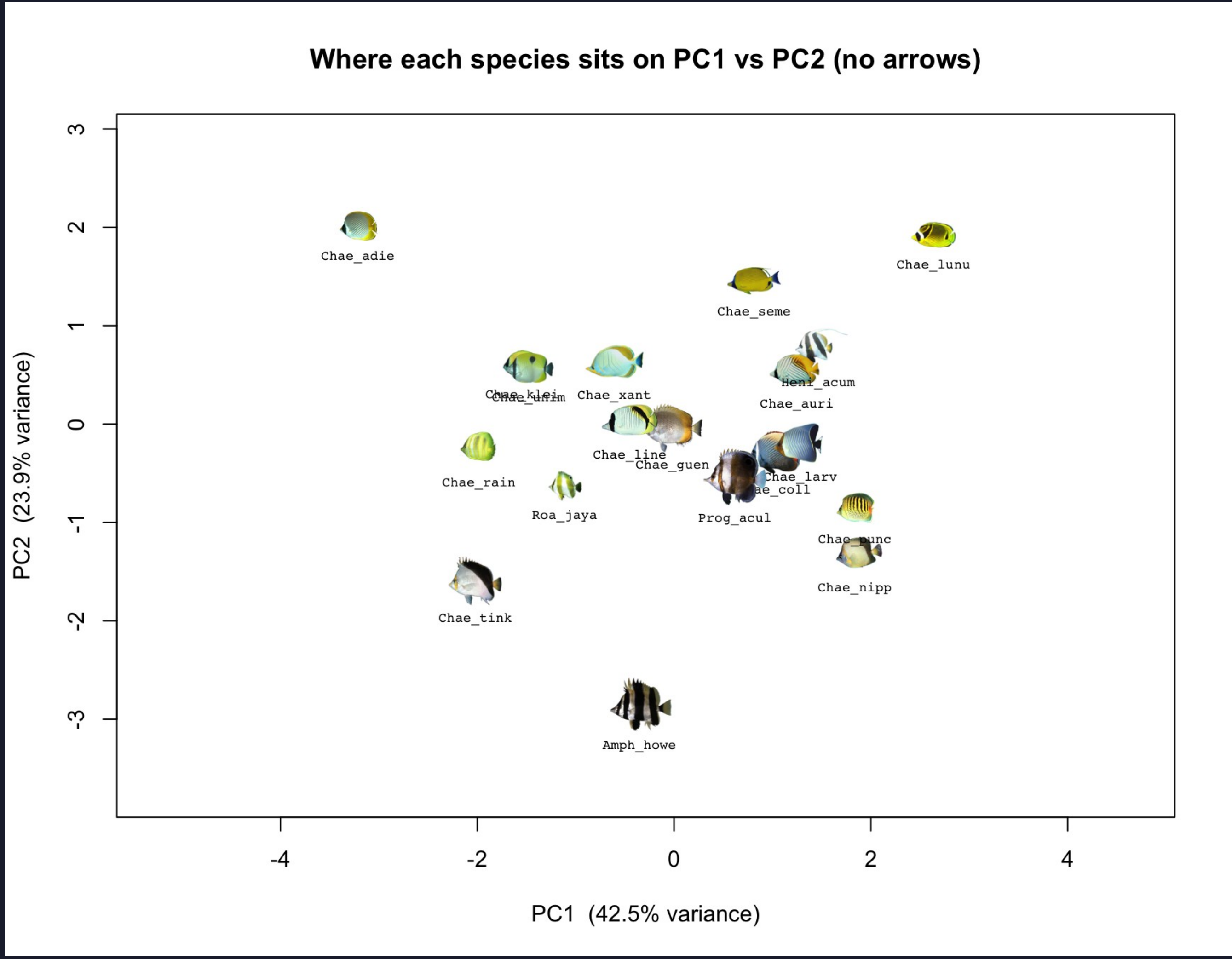
Watch for tight scatter:

redundant axes → fewer

real pattern dimensions.

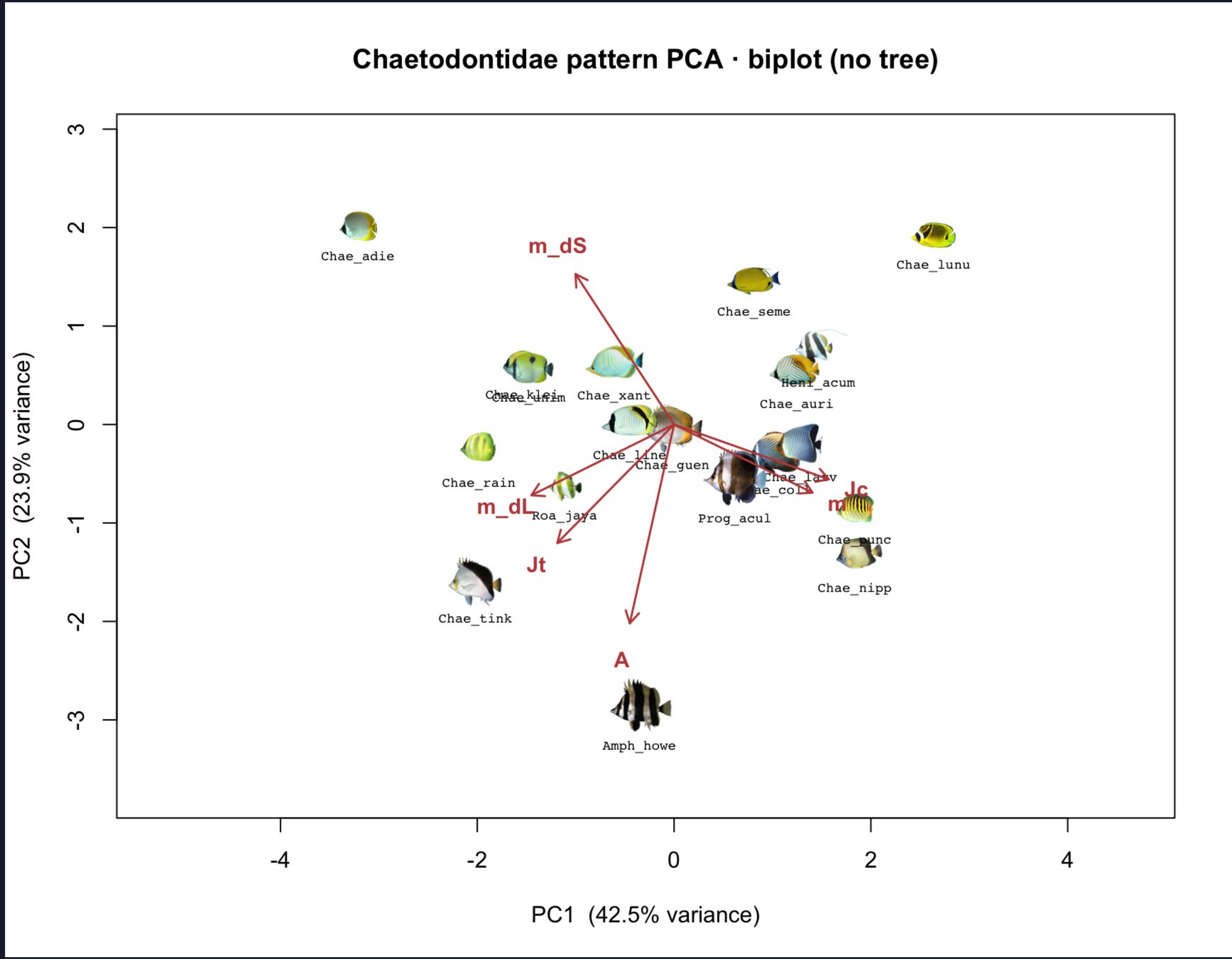
PC1 vs PC2 — where each species sits

PC1 ≈ 46 % · PC2 ≈ 17 % of pattern variation



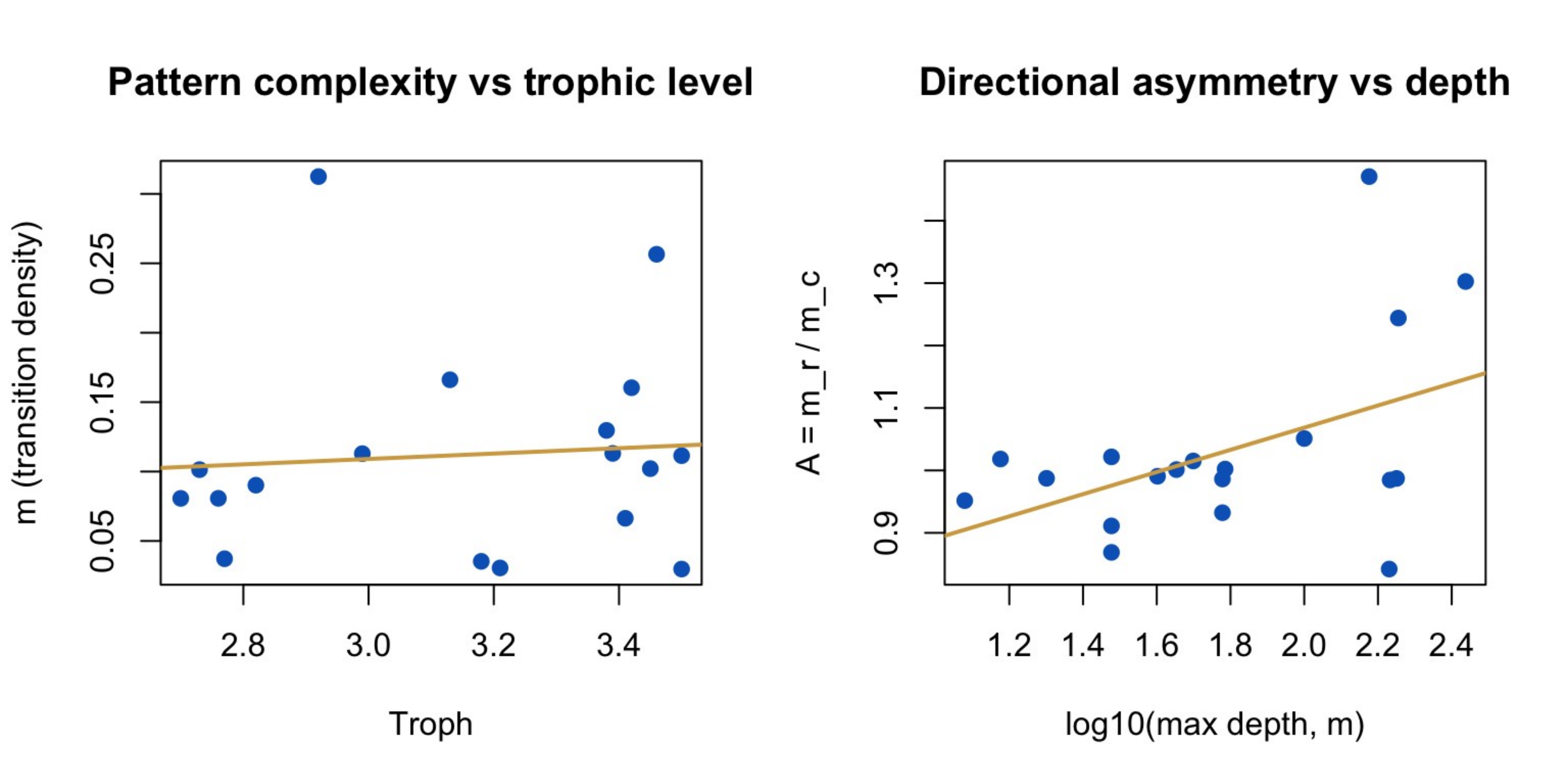
What drives the two axes

arrows = pavo variables · long arrow = high contribution



Does pattern track how the fish lives?

two named bivariate tests · ahistorical



Trophic level → complexity (m)

Depth → asymmetry (A)

Part 2 will redo these with the tree.

Now you build your bundle

15 min on FishView · same login as Demo 7

Pick ≥ 15 species across ≥ 5 genera from your family

Star one exemplar image per species

Mark ONE species as the error-species → tick 5 images for it

Hit Export. Drop the ZIP next to your Chaetodontidae bundle.

The choices you make here ARE the analysis.

Pick the species and the photos you'd defend in a paper.

Part 2 — bring the tree back

Closely related species share trait values because they share genes.

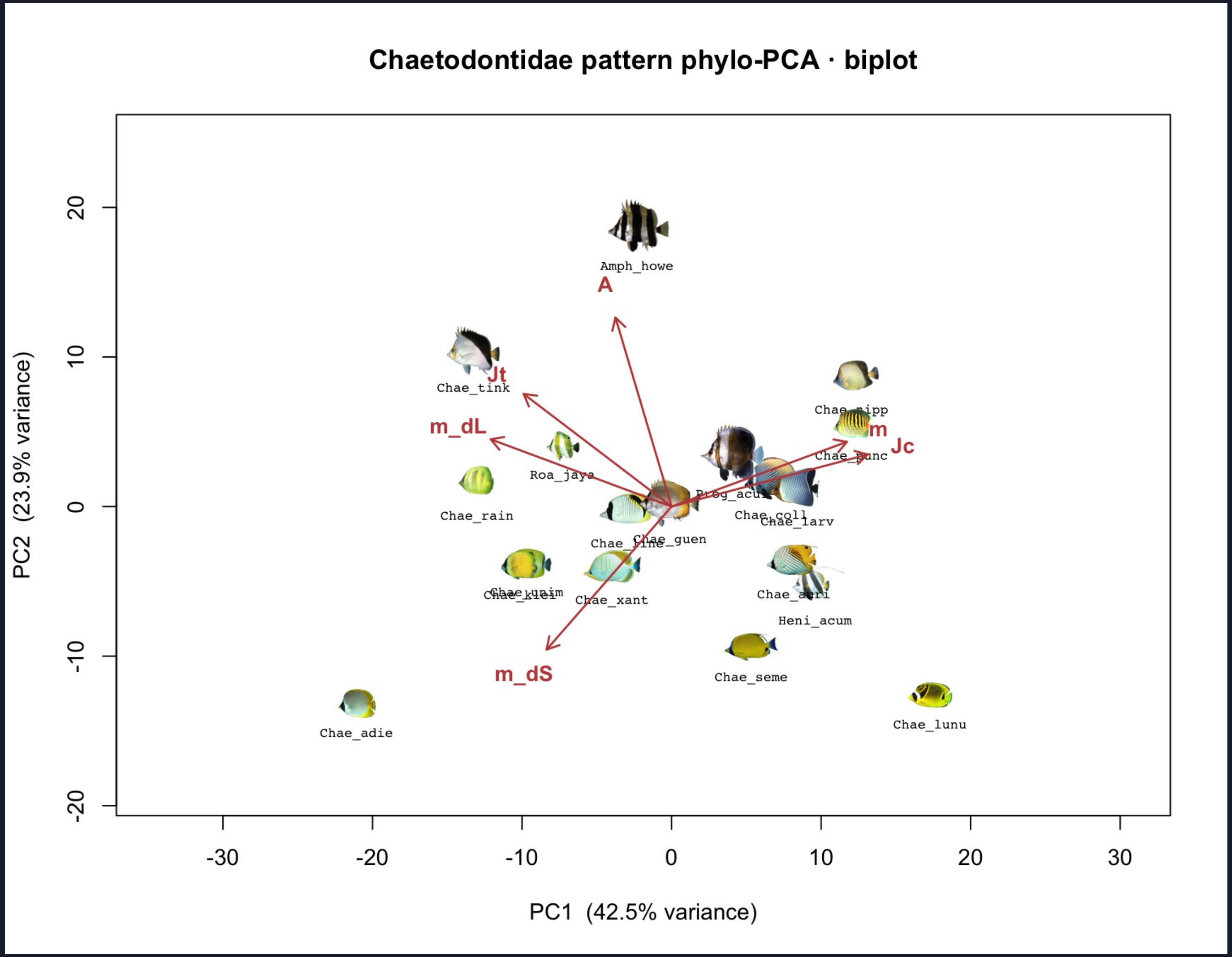
Three corrections, in order:

- phyl.pca — PCA reweighted by the tree's variance-covariance
- PGLS — bivariate regression where residuals can correlate via the tree
- K and λ — phylogenetic signal, per variable

Same questions as Part 1. Different answer if phylogeny was driving things.

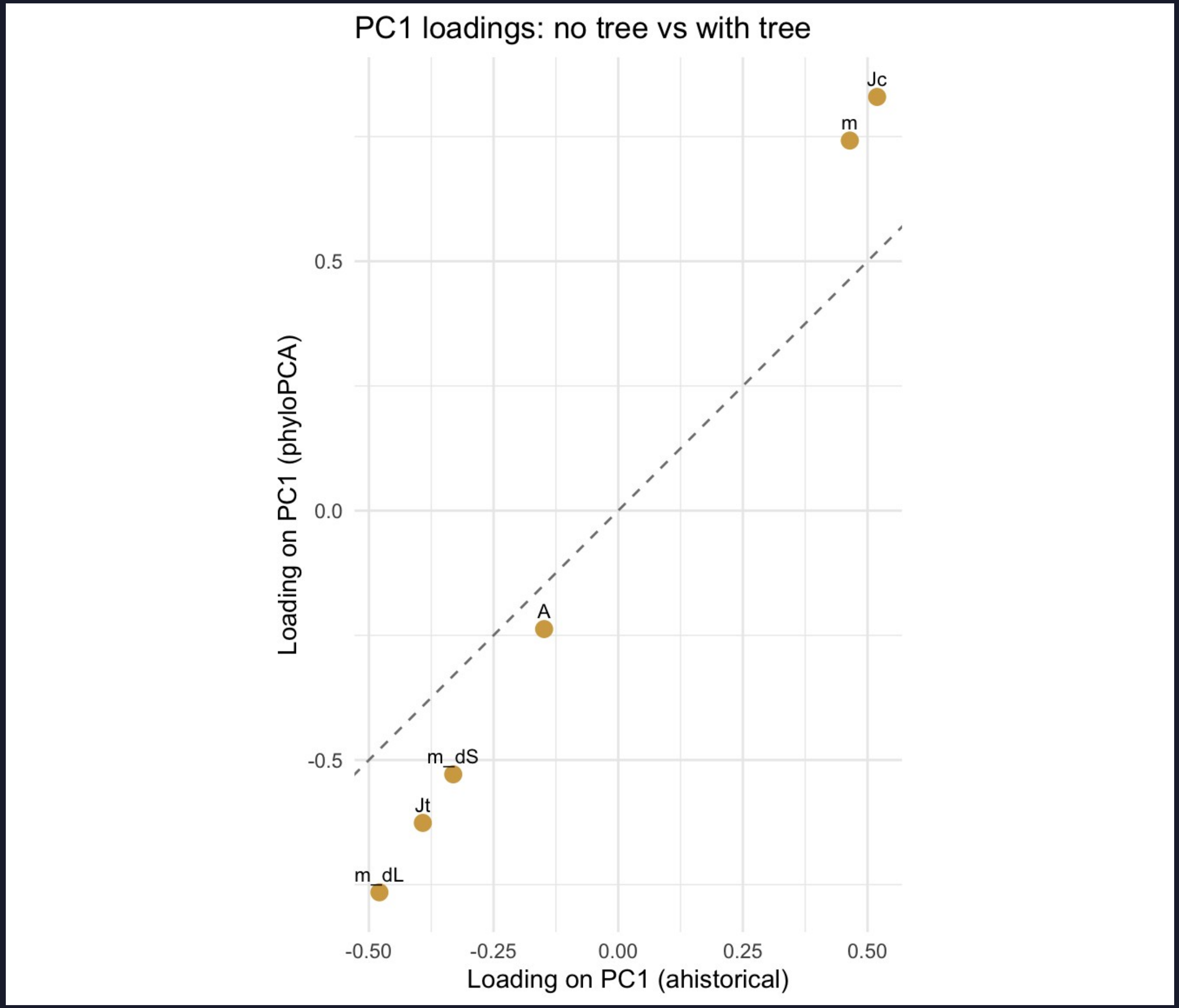
phyloPCA — same axes, different math

same 6 variables · centering uses the tree's BM variance-covariance



Did phylogeny change which variables matter?

PC1 loading: phyloPCA vs ahistorical

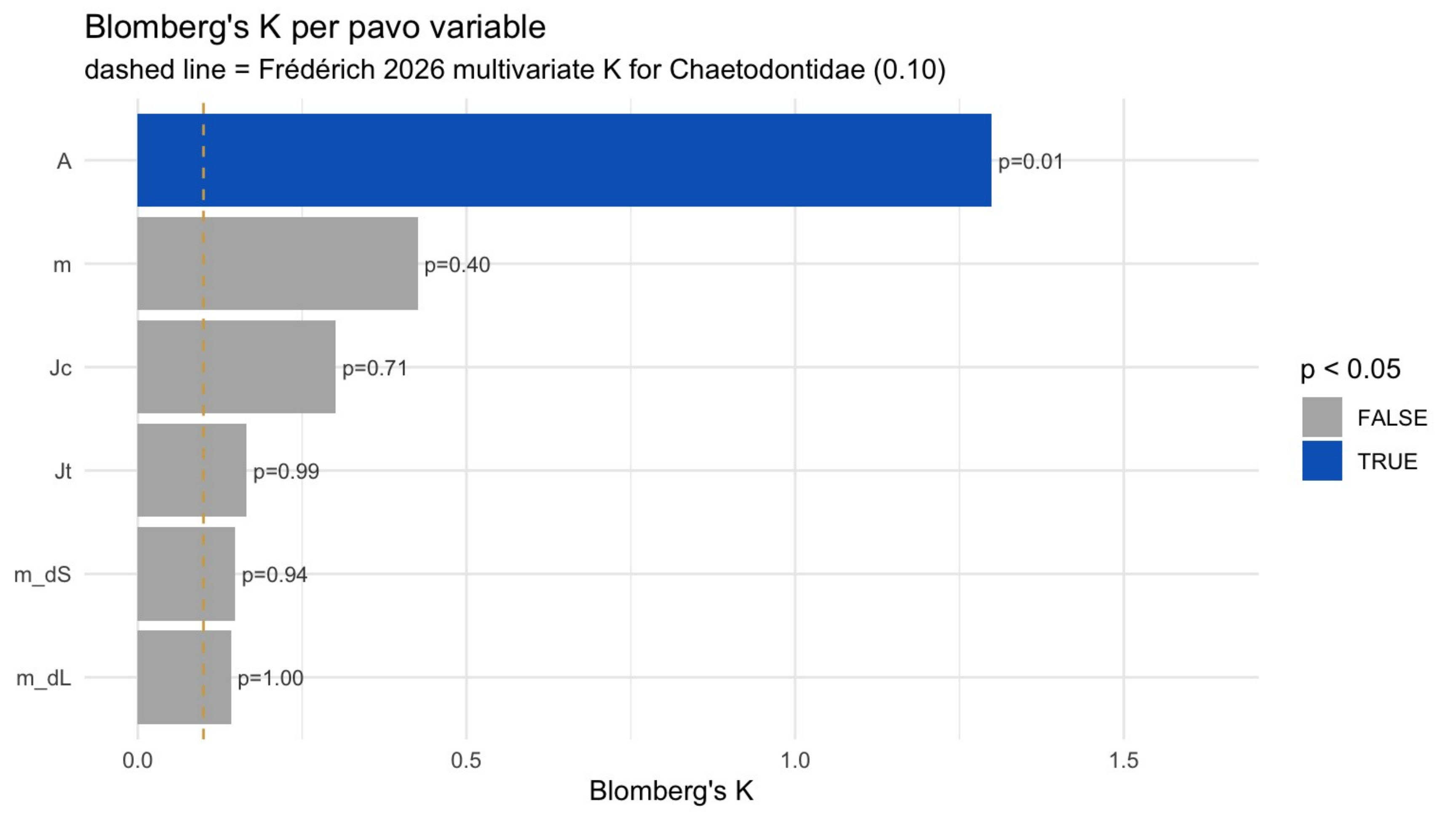


On the dashed line: phylogeny didn't change that variable's loading.

Far from it: shared ancestry was inflating it.

Blomberg's K per pavo variable

K < 1 = relatives more different than Brownian expectation

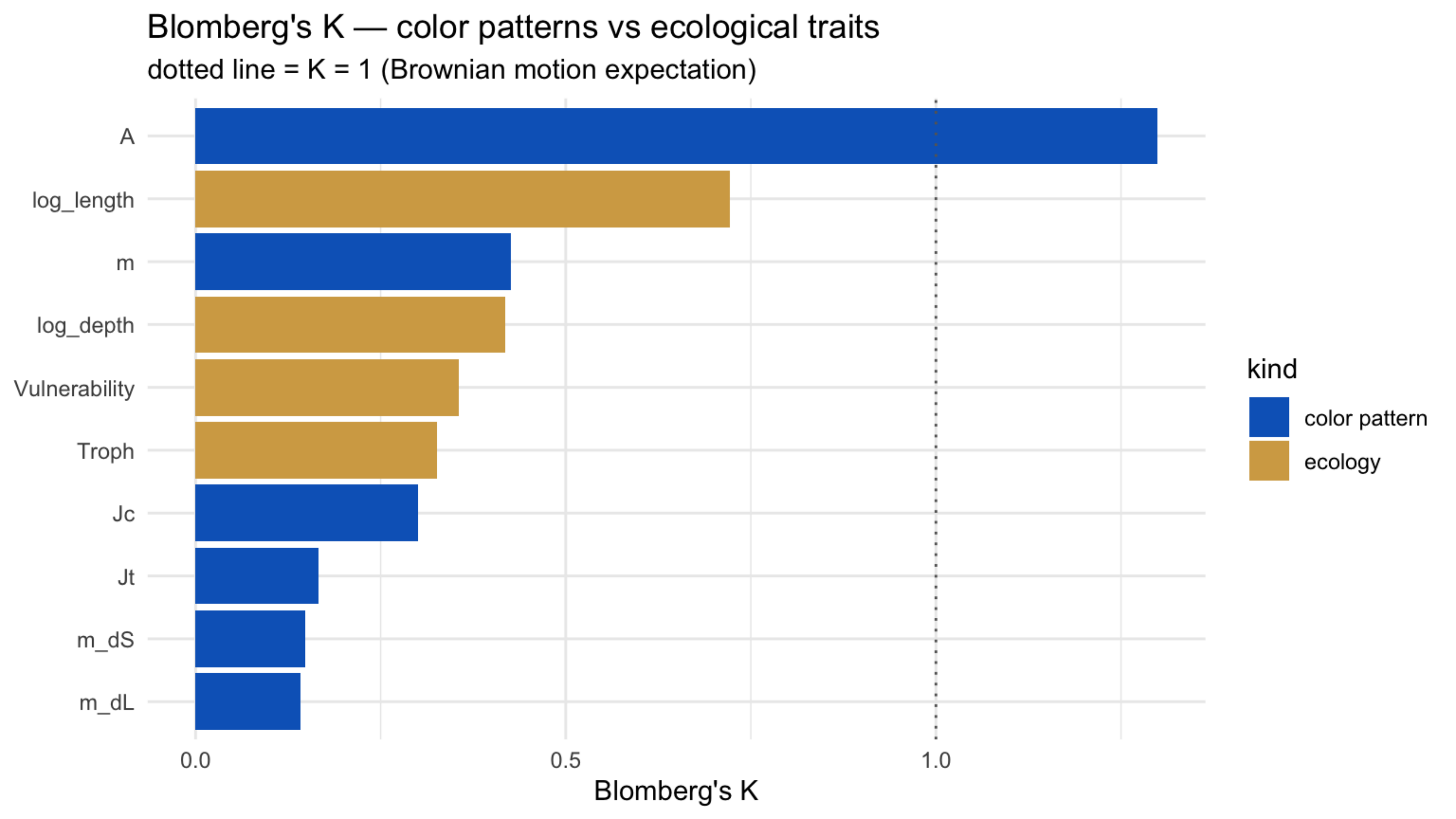


Blue bars: $p < 0.05$ — real signal.

Dashed line: Frédérich's $K = 0.10$ as a landmark, not a target.

Color signal vs ecology signal

where the demo lands



Dotted line: K = 1, BM expectation.

Watch for ecology bars above color bars in your own family — Chaetodontidae is too small to see it sharply.

Part 3 — admit you got the numbers from a photograph

Pick one species. Photograph it five times.

Compute pavo on each photo.

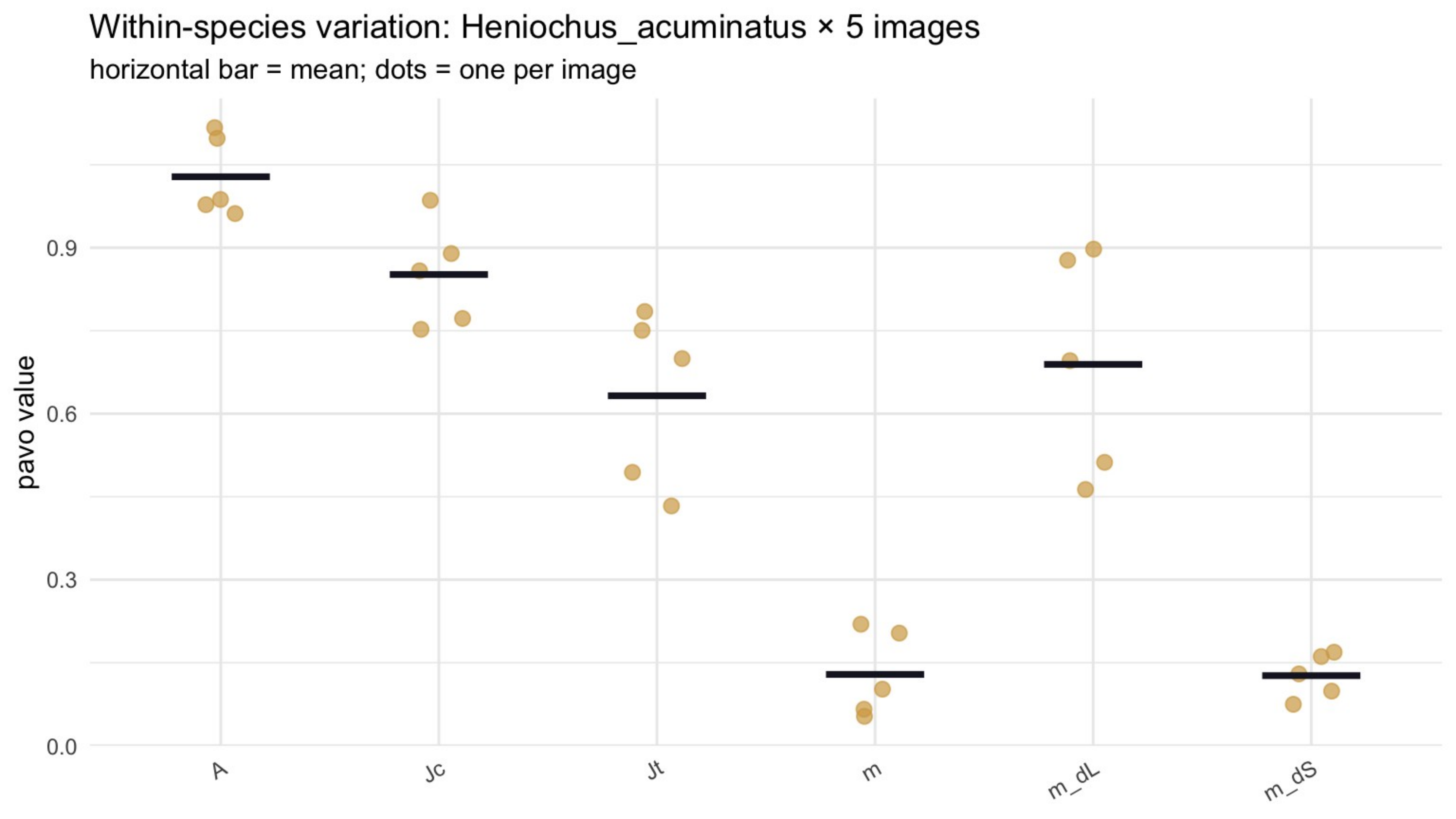
The spread is your measurement noise.

Now ask: how much of Part 2 survives noise of that size?

Monte Carlo: re-run Parts 1 & 2 a hundred times under that noise.

Within-species noise — Heniochus_acuminatus × 5 images

five dots per variable; bar = mean

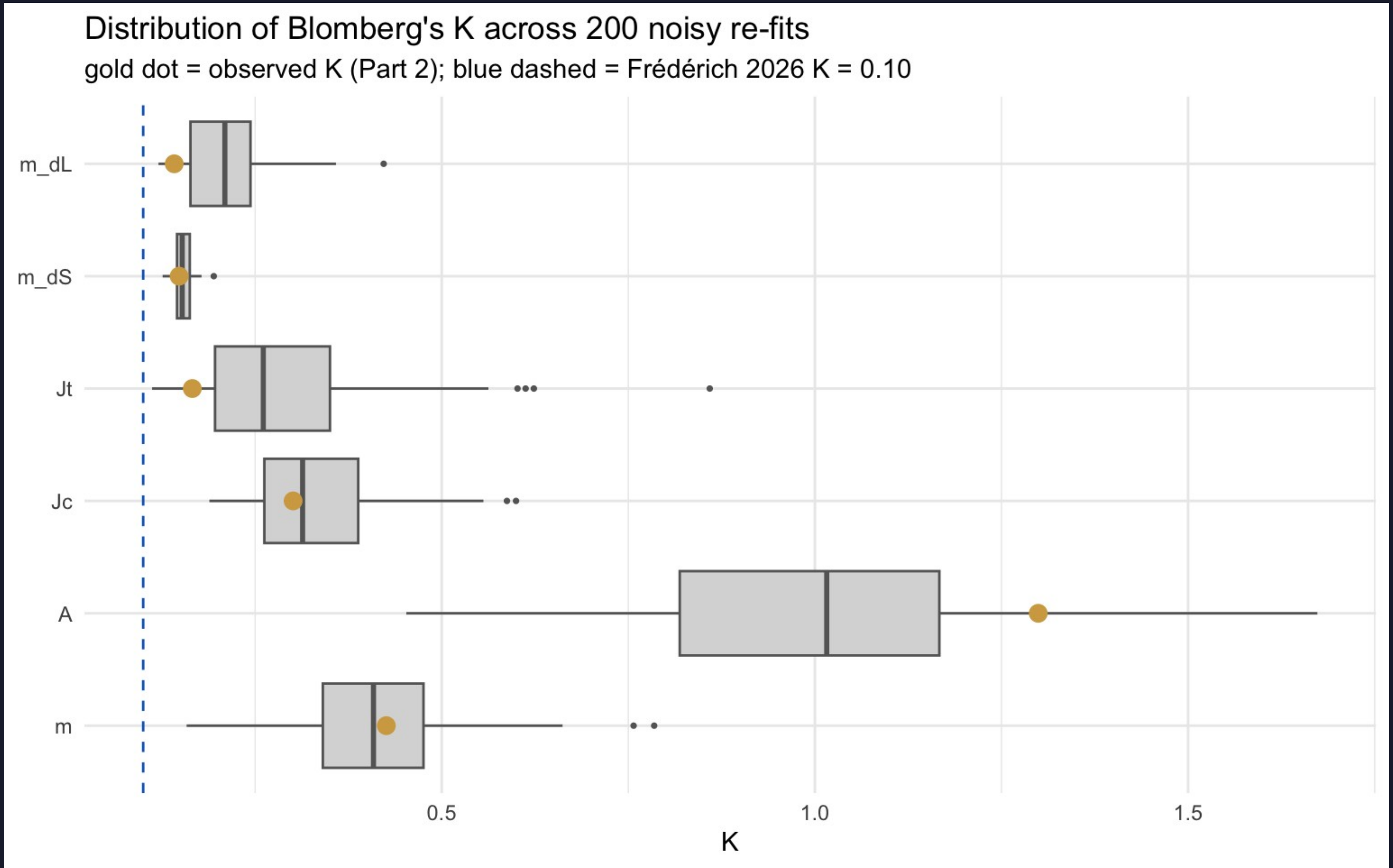


Tight clusters
→ pavo is
reliable.

Wide spread
→ that variable's
Part 2 result
is fragile.

K under noise — 100 Monte Carlo re-fits

gold dot = observed K (Part 2)



Gold inside the box: robust.

Gold outside: you got lucky — a different photo would have given a different K.

What you turn in

BruinLearn · end of day Wednesday

1. Part 1 PCA scatter — your family
2. Part 2 K plot + one-line interpretation per variable
3. Part 3 Monte-Carlo CI on one statistic of your choosing
4. One paragraph (≤ 250 words): does color in your family match Frédérich, and which conclusions survive measurement error?

Tonight

prep page · ~20 minutes

1. Install R packages (one chunk on the prep page)
2. Download the Chaetodontidae walkthrough bundle
3. Run the readiness check until it prints PASS
4. Confirm your Demo 7 FishView login still works — you'll build your bundle in class

michaelalfaro.github.io/eeb187-demos/demo8/demo8-prep.html

See you Wednesday.

Bring your laptop charged. We move fast.